

Intrusion Forecasting Framework for Early Warning System against Cyber Attack

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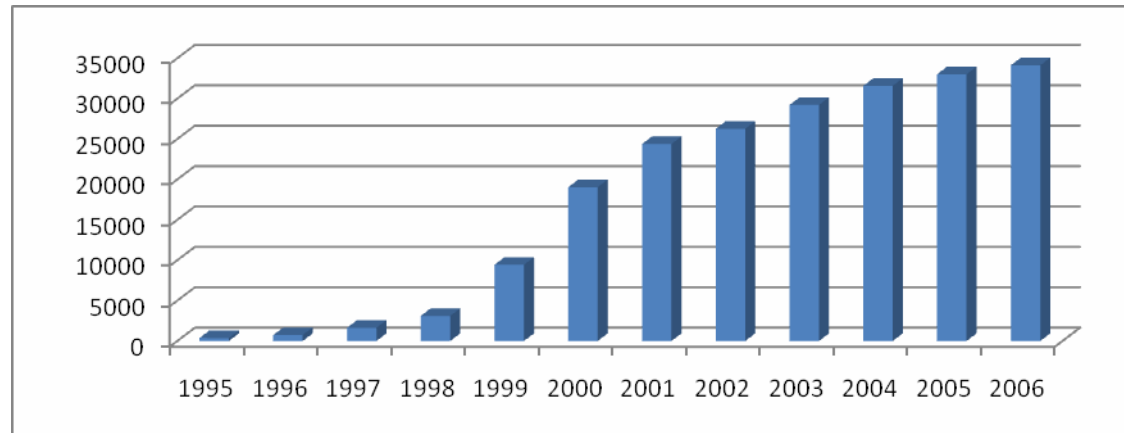
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- 1 Recent Cyber Attacks
- 2 Early Warning System
- 3 Intrusion Forecasting System
- 4 Conclusion

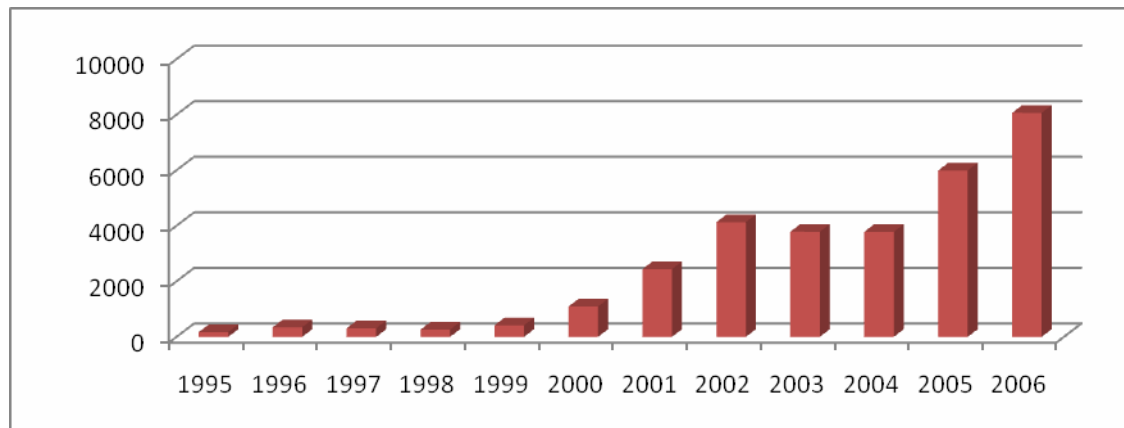
-  **Recent Cyber Attacks**
-  **Early Warning System**
-  **Intrusion Forecasting System**
-  **Conclusion**

Growth of the Internet usage

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Number of Internet users in Korea (unit: 1,000) (Source: isis.nida.or.kr)

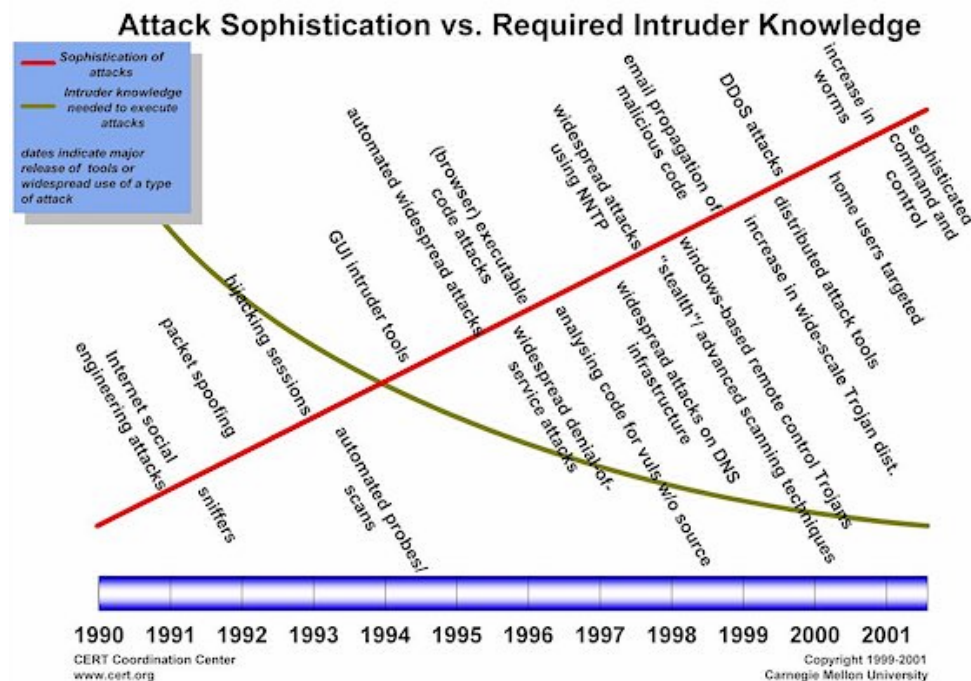


Number of vulnerabilities identified (Source: www.cert.org)

Attack Trend

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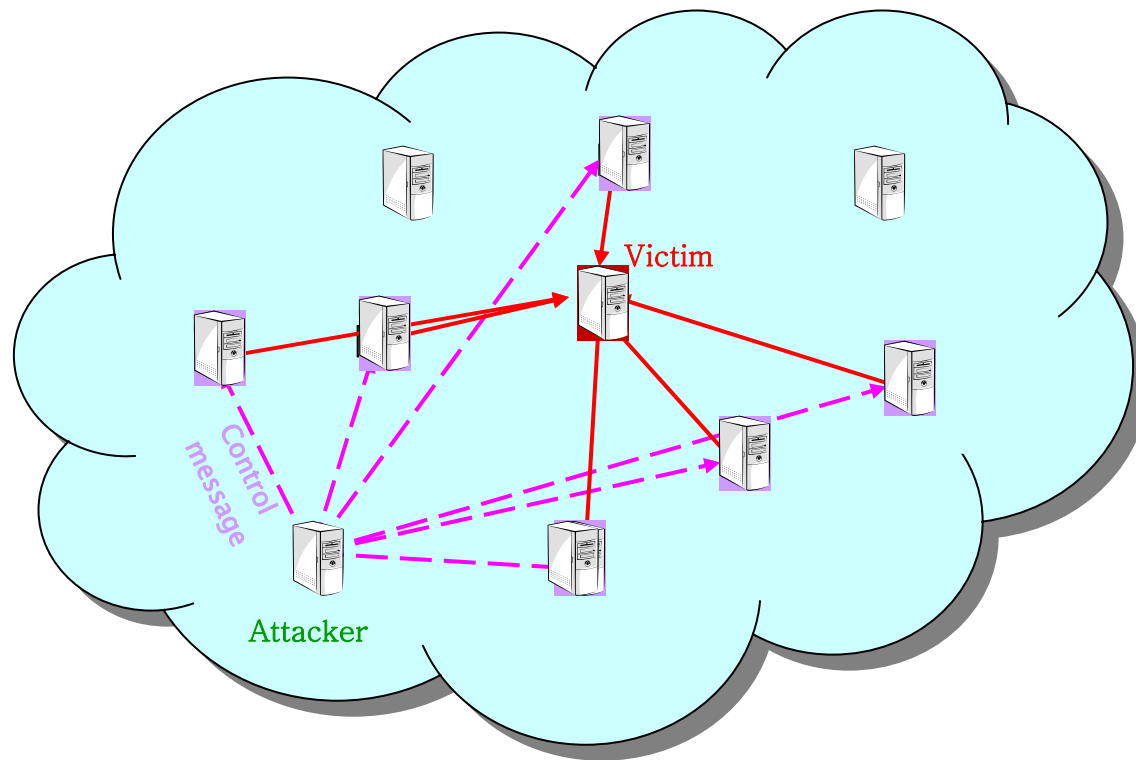
- Exploit the interconnectivity of networks
- Rapid attacks, sometimes zero-day
- More sophisticated and evolutionary attack tools
- Attacks on infrastructure



Source: www.cert.org

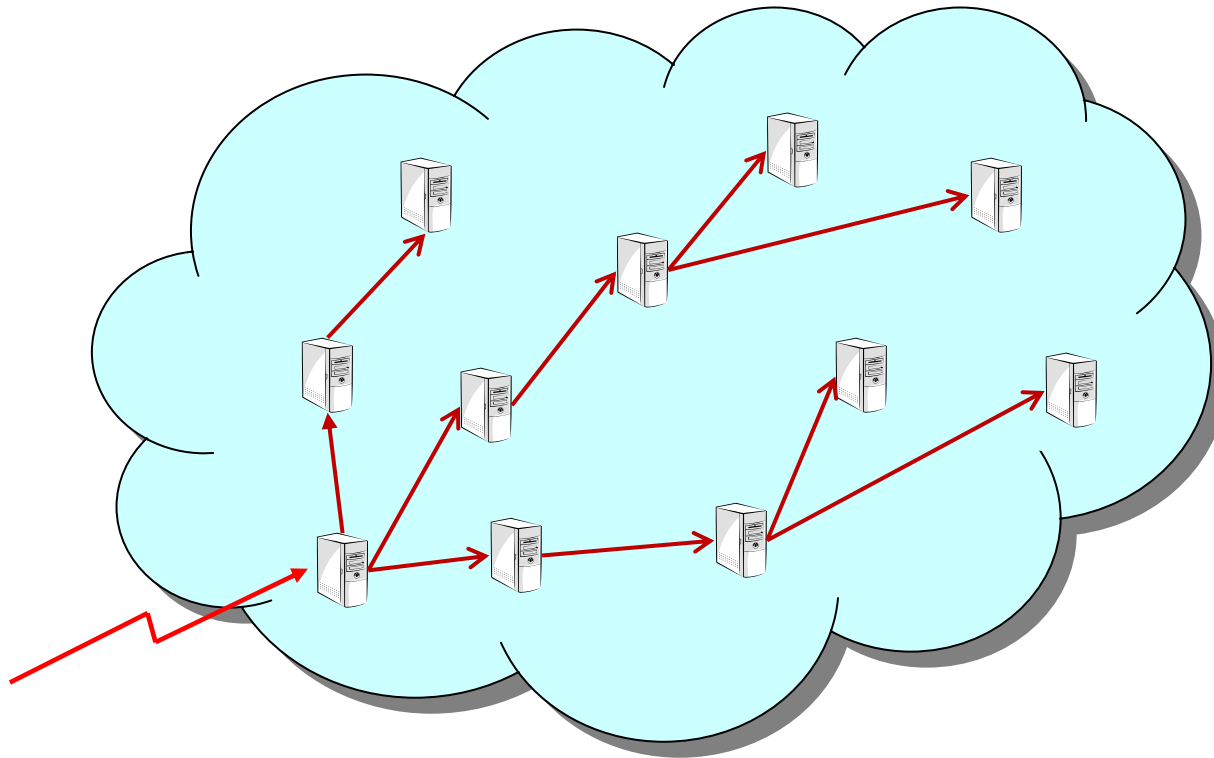
DDoS Attacks

- Deploy a large number of compromised systems to attack a victim host



Internet Worm

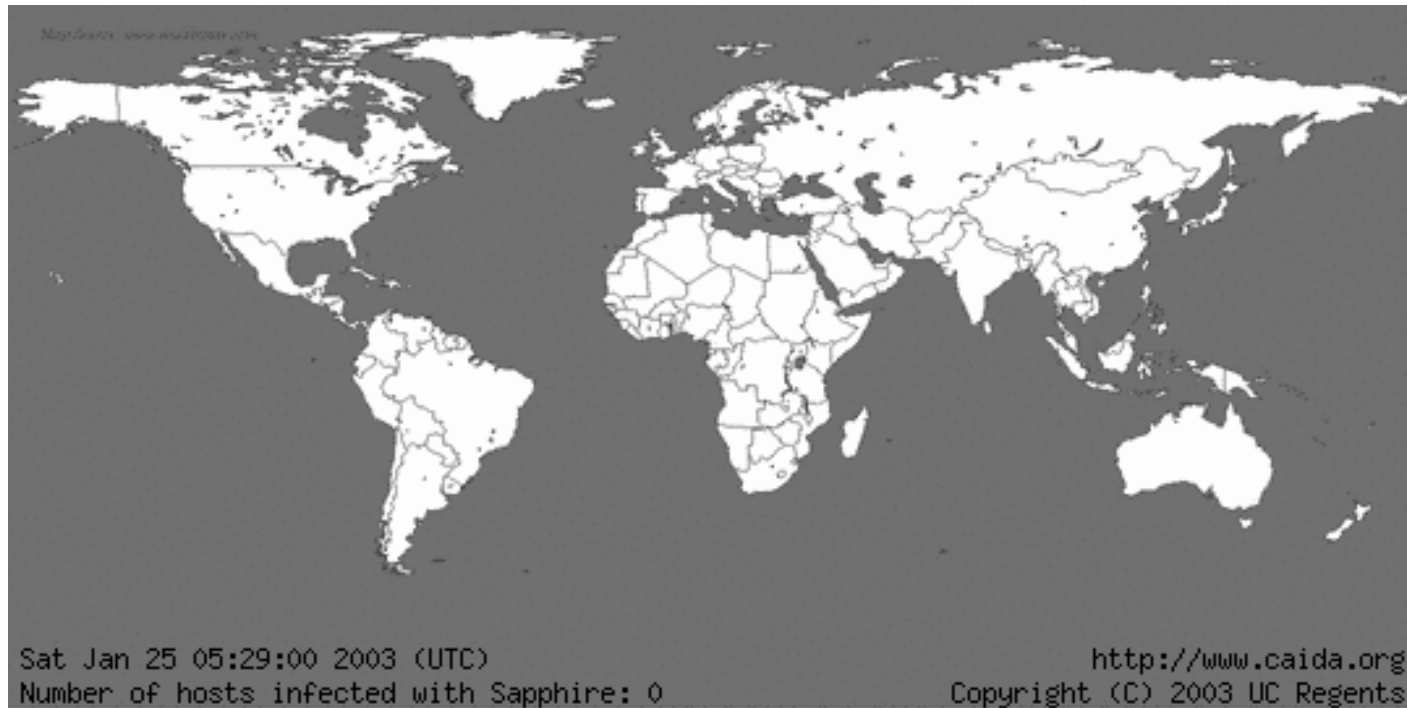
- Self-propagating & self-replicating network program



Internet Worm

■ Slammer worm

- Infected more than 90% of vulnerable hosts within 10 mins.
- Caused shutdown of Internet service in Korea.



The number of infected slammer hosts in the 30 minutes after release (Source: www.caida.org)

Botnets

■ Bot

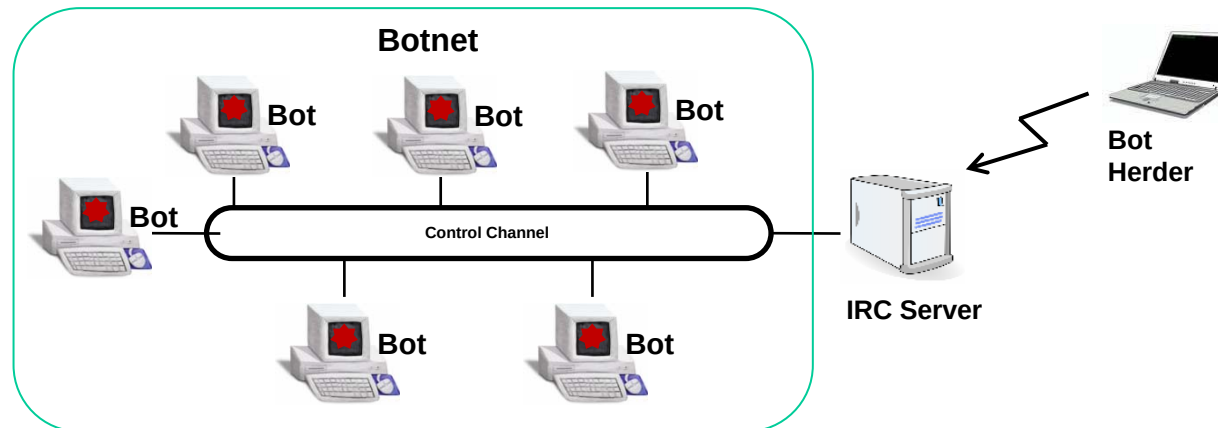
- A short word for “robot”
- Piece of software that allows a system to be remotely controlled

■ Zombie

- Controlled/corrupted system

■ Botnet

- A network of Zombie systems
- DDoS, Spamming, Sniffing, Key Logging, Identity Theft, Hosting of Illegal Software



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Early Warning System

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■ What is EWS?

- A system or procedure designed to warn of a potential or an impending problem in order to minimize the damage against problem
- More important as the damage becomes more tremendous

■ EWS in Real Life

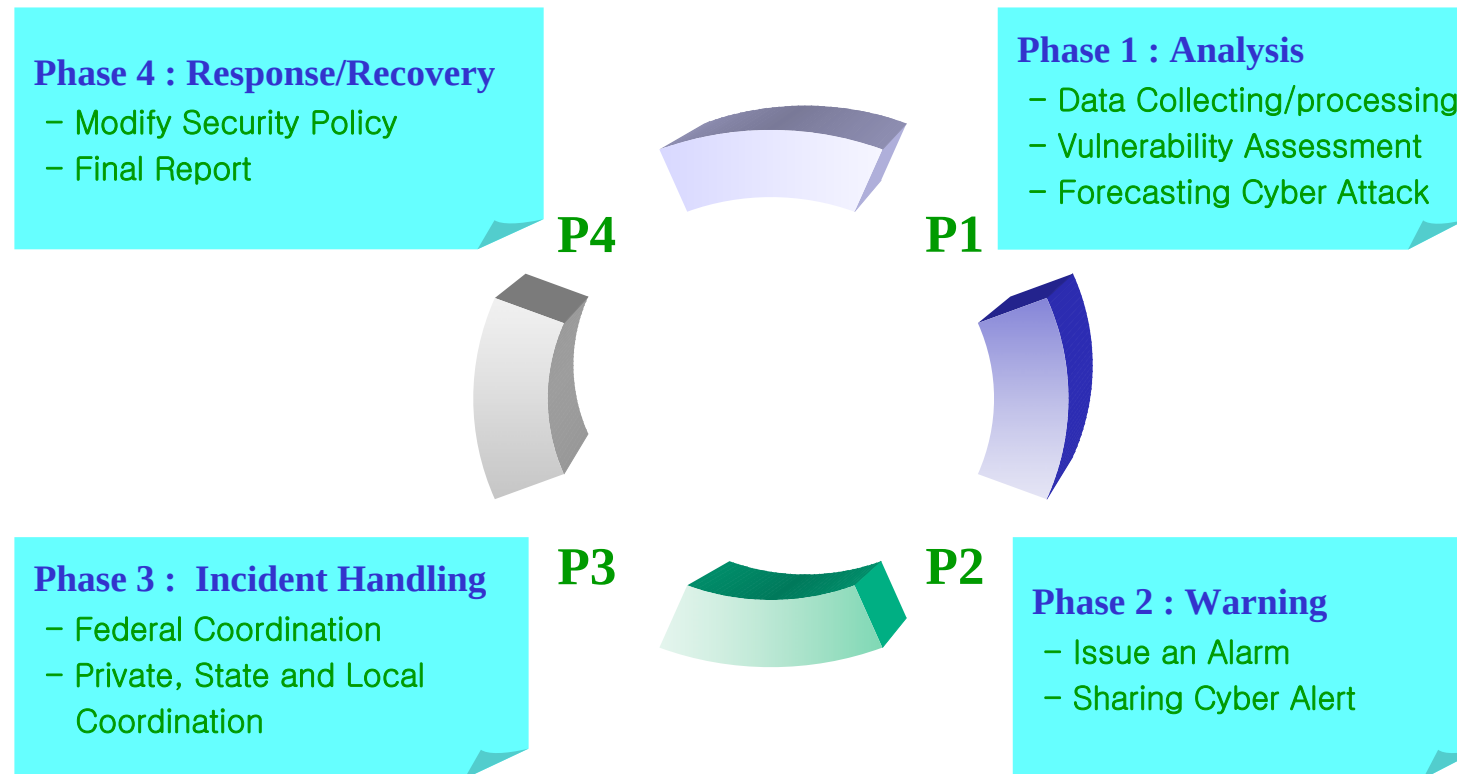
- Famine EWS
- Disaster (Tsunami/Earthquake) EWS
- Disease EWS

■ EWS in Cyber Space

- Early detection of Cyber Attack and Instant response to it
- Cyber Attack can be Intrusion, Worm/Virus Outbreak, Information Warfare and so on.

EWS Procedure

The main procedure of EWS conforms to that of *National Cyberspace Security Response System* (USA)



■ Computer Emergency Response Team

- Coordinate all of the activities of organizations and institutions involved in efforts to **secure national IT network**
- Protect public/national security from cyber threats **by handling computer incidents promptly and efficiently**
- Traditionally, **EWS is operated by CERT**



■ Motivation

- 'Basic Plan for the Establishment of National Cyber Terror Response System' is approved by President, July 2003

■ Related Organization

- NCSC (National Cyber Security Center)
 - ✓ Central point of government for identifying, preventing and responding to cyber attack and threats in Korea
- KISA (Korea Information Security Agency)
 - ✓ Agency providing public user, industries and organization with information security service
- CERT
 - ✓ KrCERT (Korea CERT) : CERT operated by KISA
 - ✓ KN-CERT (Korea National CERT) : CERT operated by NCSC

■ Motivation

- FISMA : Federal Information Security Management Act of 2002
- Planning 'National Strategy to Secure Cyberspace', Feb. 2003

■ Related Organization

- DHS/NCSD (National Cyber Security Division)
 - ✓ Division that works to secure cyberspace and America's cyber assets.
 - ✓ Within DHS (Department of Homeland Security)
- CERT/CC (CERT/Coordination Center)
 - ✓ CERT Center operated by Carnegie Mellon University
- US-CERT (U.S. Computer Emergency Readiness Team)
 - ✓ US-CERT is charged with protecting USA's Internet infrastructure by coordinating defense against and response to cyber attacks
 - ✓ Founded by CERT/CC & NCSD

■ Motivation

- Decree 2001-693, July 2001 : Organize DCSSI
- 'State Information System Security Reinforcement Plan' , 2004

■ Related Organization

- SGDN (Secrétariat général de la défense rationale)
 - ✓ By Decree 96-67, SGDN takes charge of Cyber Security of France
 - ✓ Belong directly to Prime Minister
- DCSSI (Direction Centrale de la Sécurité des Systèmes d'Information)
 - ✓ Execute governmental task in order to protect information system
 - ✓ Operate CERTA and ITSOC, under the authority of the SGDN
- ITSOC (IT Security Organization Center)
 - ✓ Research specialized knowledge to prevent and solve security incident
 - ✓ Collect data through CERTA and provide authorities with collected data

Research Activity in EWS

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■ Early Detection of Incident

- Incident (Intrusion) Forecasting Method
- Alert Correlation
- Threat Assessment

■ Issue of an alarm or warning

- Design of main framework for EWS
- Partnership with other related organization
- Effective visualization of event

■ Response/Recovery

- Traceback Mechanism
- Establishment of national cyber security policies or law

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Intrusion Forecasting System

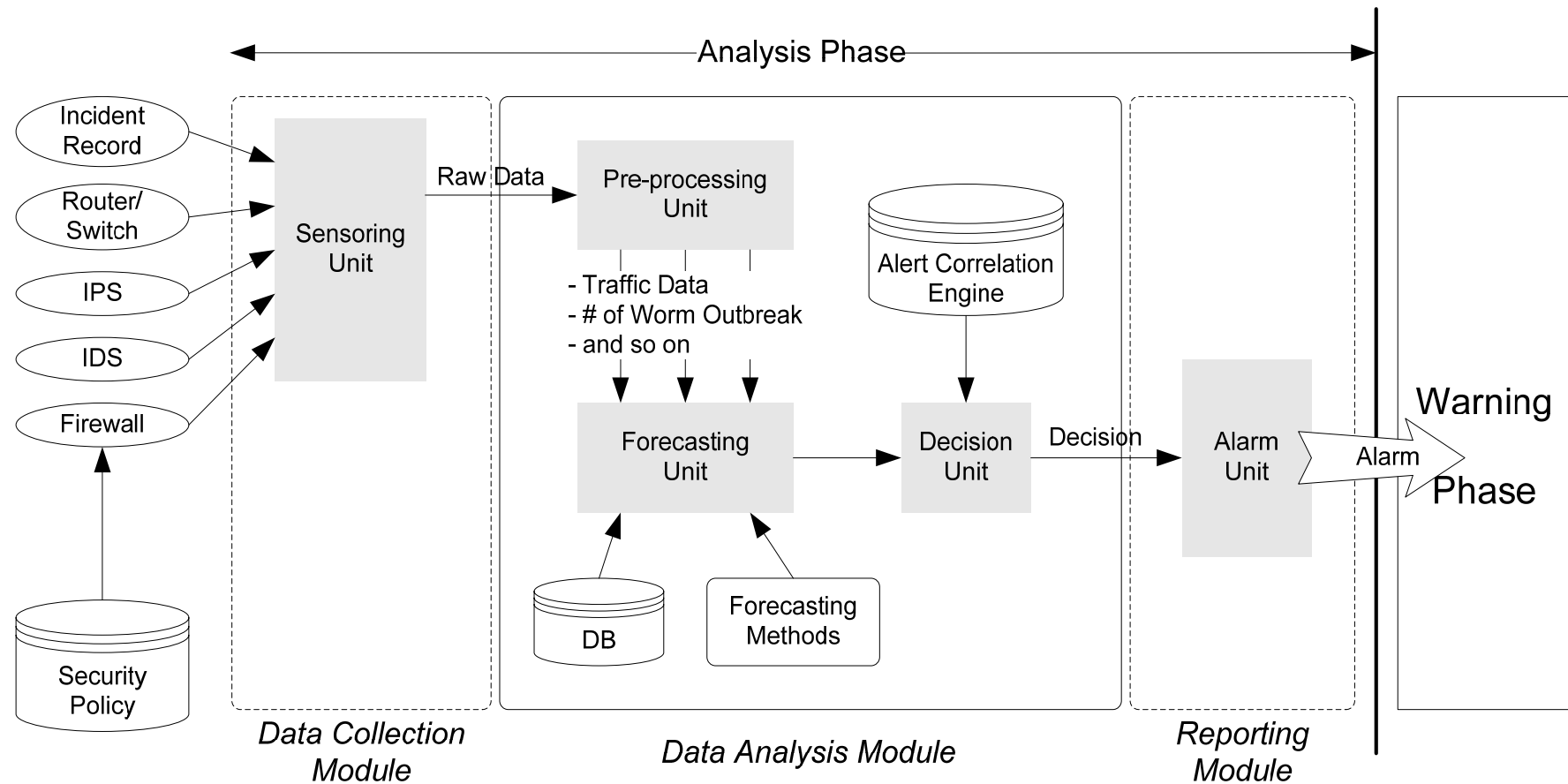
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■ Intrusion Forecasting System

- Forecast cyber attacks in advance
- The most significant part in EWS
 - ✓ Analysis phase corresponding to the *National Cyberspace Security Response System*
 - ✓ The speed and precision of detection is a key factor of EWS
- Development of an intrusion forecasting system is necessary

Intrusion Forecasting System Architecture

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Intrusion Forecasting System Architecture

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■ DC Module

- Collect data from various sensors
- Pre-processing for the DA module

■ DA Module

- Analyze the collected data
- Predict possibilities of cyber attack

■ REP Module

- Create alarm reports
- Alarm visualization

Forecasting in Real Life

Weather Forecasting Case

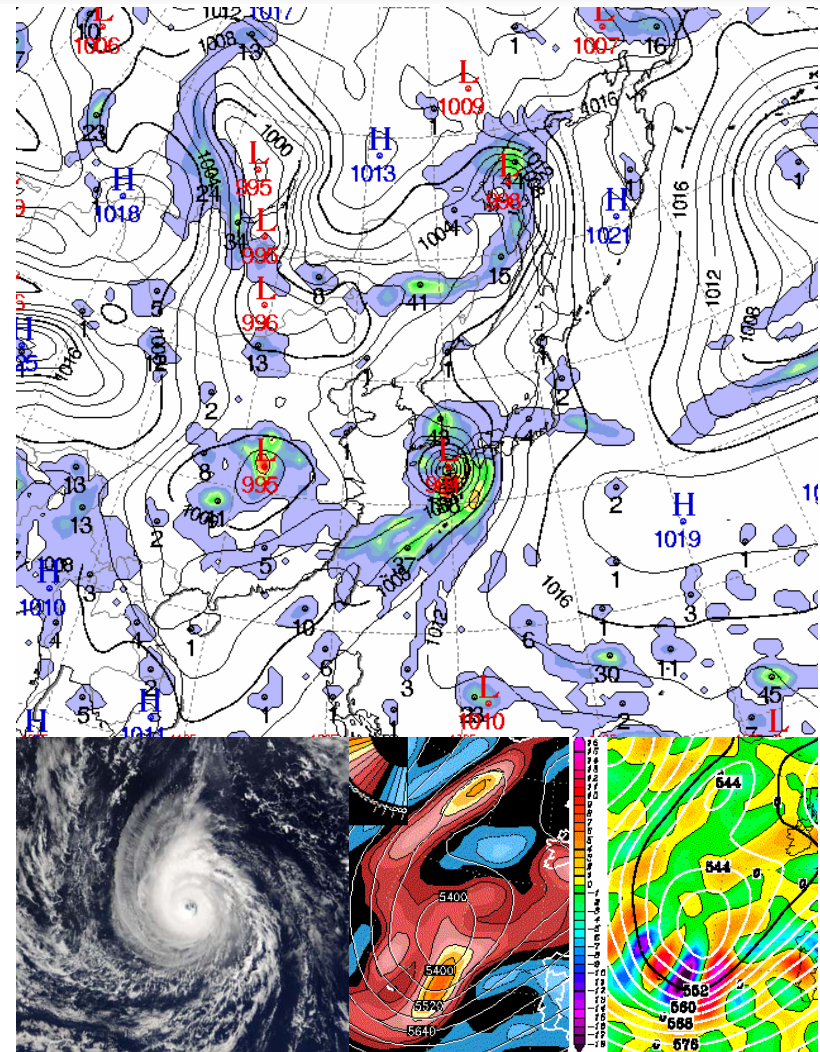
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■ Three Important Forecasting Steps

- Analyze the present state
- Predict a future state
- Interpret the model results

■ Methods of Weather forecasting

- Folklore forecast
- Persistence forecast
- Climatology forecast
- Trend forecast
- Analog forecast
- Numerical forecast
- Ensemble forecast



Intrusion Forecasting Methods

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■ Forecasting Methods against Cyber Attack

- Still at primary level compared to other forecasting areas
 - ✓ predict virus day or possibility of cyber threat by exploiting security vulnerabilities
 - ✓ Commercial intrusion forecasting system is currently in beginning phase
 - ✓ Some researches have applied existing forecasting techniques to the prediction of worms or viruses

- Related organization
 - ✓ Warning virus/malicious code : Ahnlab, Hauri, SANS (Internet Storm Center)
 - ✓ Threat management system : Symantec (DeepSight TMS), Computer Associated (eTrust Security Management)

Intrusion Forecasting Method

(1) Data Mining Method

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■ Data Mining Method

- Extract implicit, previously unknown, and potentially useful information from large data sets or databases
- Widely used in the various forecasting areas
 - ✓ Stock prices, weather forecasting, and earthquake forecasting
- Possible to handle numerous traffic variables difficult to analyze intuitively
- Clustering Analysis, Decision Tree, Genetic Algorithm, etc.

■ Advantages vs. Disadvantages

- Advantages
 - ✓ Consider not only quantitative changes of multiple variables but also changes of their distribution
- Disadvantages
 - ✓ High computational complexity
 - ✓ Difficult to understand the results

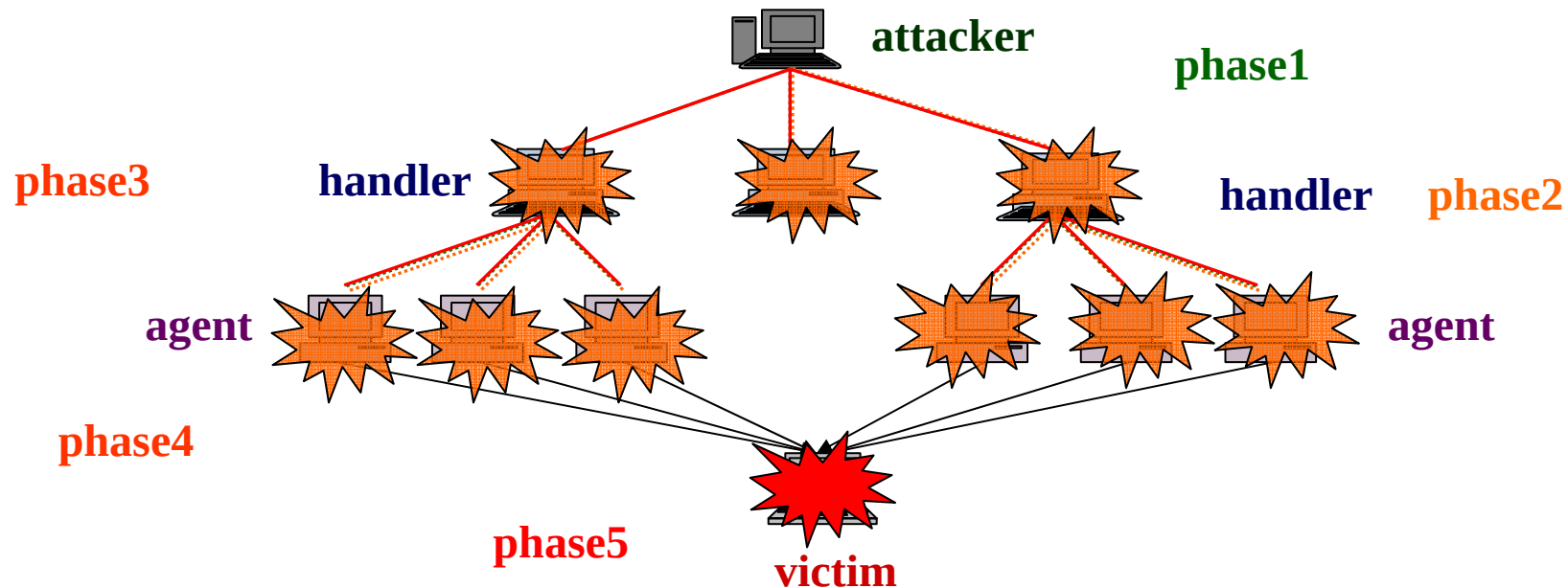
A CASE STUDY : Data Mining Method

DDoS attack detection method using cluster analysis

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■ A Proactive detection of DDoS attack

- Detect DDoS attack proactively by exploiting the sequential movement of DDoS
- Attack phases of DDoS attack



A CASE STUDY : Data Mining Method

DDOS attack detection method using cluster analysis

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■ Feature selection process

- Select several features to detect the symptoms of a DDoS attack
- Indicate abnormal changes in traffic according to each phase of the attack.
- Selected features
 - ✓ Distribution of source/IP port
 - ✓ Distribution of destination IP/port
 - ✓ Packet type
 - ✓ Occurrence rate of TCP SYN, UDP, ICMP
- The entropy values of the selected features are calculated to measure the randomness in their distribution.
- Entropy

$$H = -\sum_{i=1}^n P_i \log_2 P_i$$

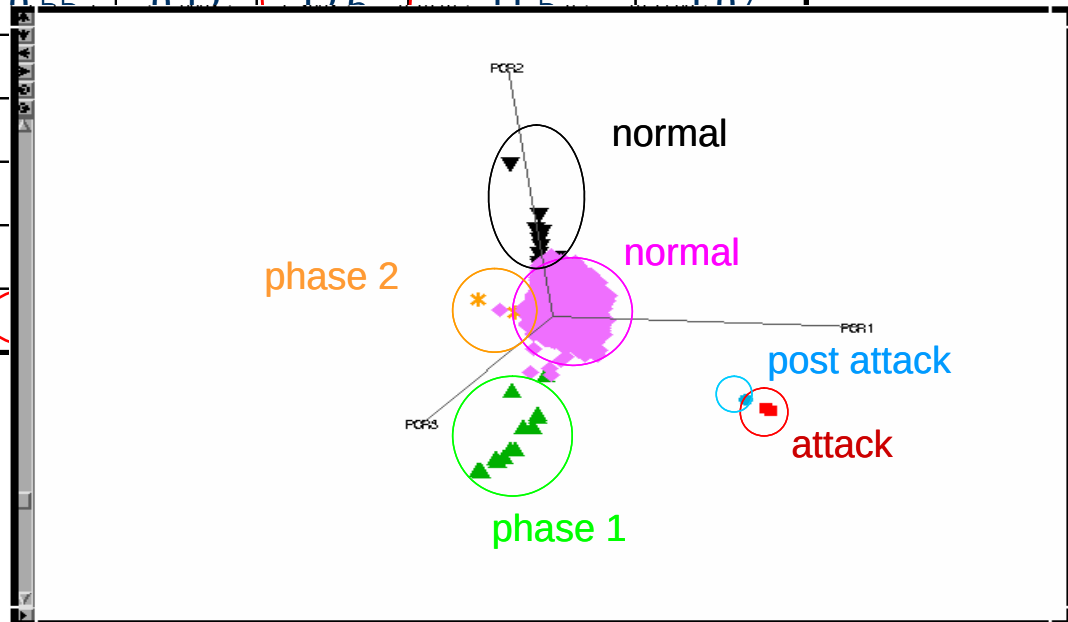
A CASE STUDY : Data Mining Method

DDOS attack detection method using cluster analysis

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■ Results of clustering analysis

cluster variable	1 normal	2 ph1	3 ph 2	4 attack	5 post attack	6 normal
Entropy of src IP	1.59	0.71	0.08	0.02	0.13	1.06
Entropy of src port	1.61	0.56	0.12	12.4	11.4	1.07
Entropy of dest IP	1.58	4.91	0.07	12.6	11.5	1.06
Entropy of dest port	1.50	0.55	0.12	12.6	11.5	1.07
Entropy of packet type	1.12					
Number of packet	37.0					
Occurrence rate of SYN	0.02					
Occurrence rate of UDP	0.00					
Occurrence rate of ICMP	0.00					



Intrusion Forecasting Method

(2) Probabilistic Modeling

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■ Probabilistic Modeling

- Capture evidence of intrusions in terms of a probability from the current network state
- Enable system administrator to understand the degree of risk on a probabilistic scale
- Markov chain, Bayesian method, etc.

■ Advantages vs. Disadvantages

- Advantages
 - ✓ Easy to understand the possibility of attacks based on a probabilistic scale
 - ✓ Highly applicable in the determination of the warning level
- Disadvantages
 - ✓ Difficult to construct the state profile and transition probabilities between them
 - ✓ Require correct decision of a system administrator

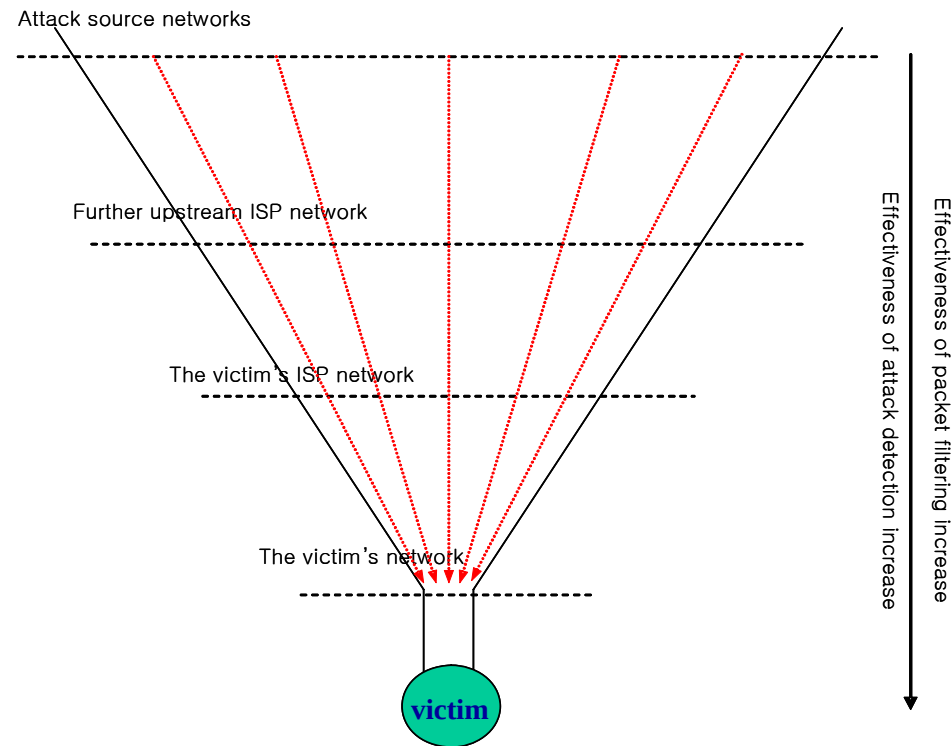
A CASE STUDY : Probabilistic Modeling

An effective DDOS attack detection and packet-filtering scheme

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■ Effectiveness of DDoS attacks Detection & Filtering

- DDoS attacks are viewed as congestion event in routers
- Effectively detected at the victim network, but effectively filtered when closer to the attack source



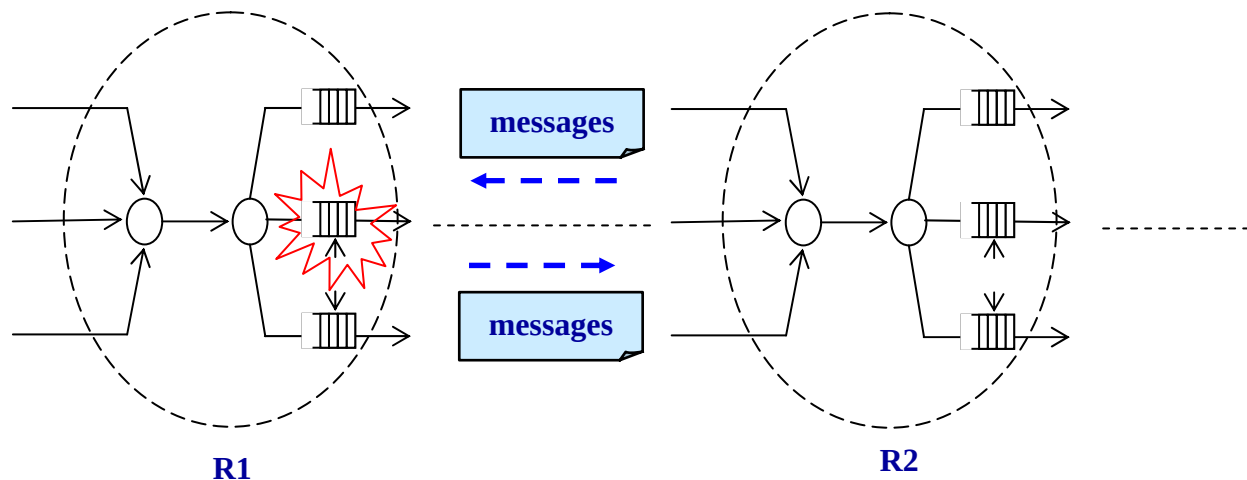
A CASE STUDY : Probabilistic Modeling

An effective DDOS attack detection and packet-filtering scheme

Telecommunication System & Internet Security Lab.

■ Measure for deciding congestion level in a congestion router

- Decide congestion level using packet loss probabilities
 - ✓ Congestion occurred at an output queue in a transit router if packet loss probability is larger than given threshold
- Detect attacks in routers through the use of queueing model



A CASE STUDY : Probabilistic Modeling

An effective DDOS attack detection and packet-filtering scheme

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■ Detection & Filtering strategy

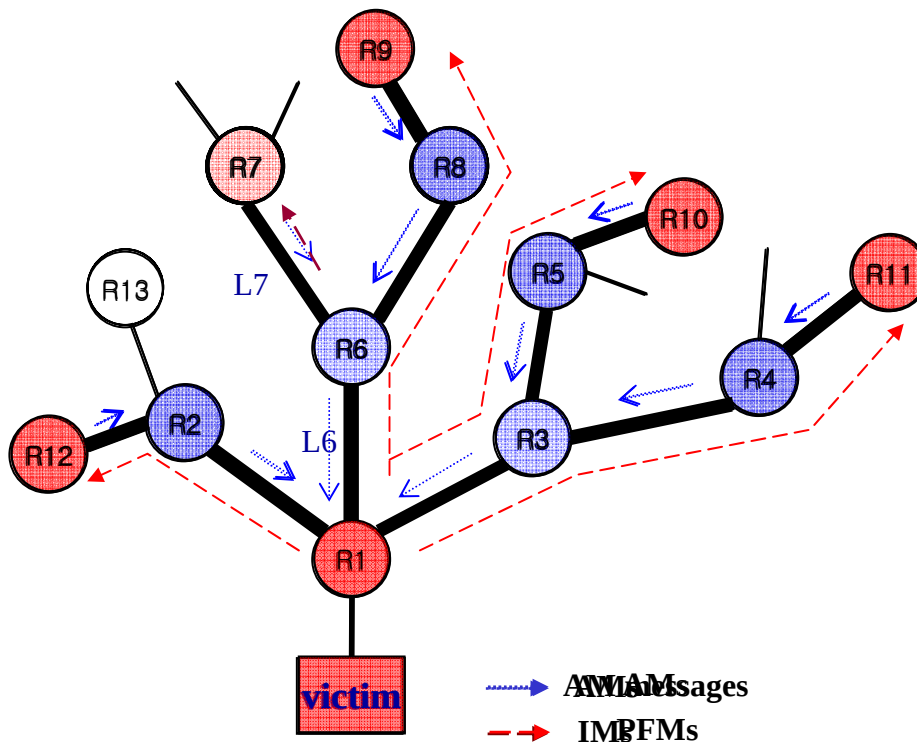
- Detect attacks in routers through the use of queueing models
- Perform congestion control in consideration of packet loss probabilities in routers
- Local & global detection by exchanging congestion messages

✓ In local detection

- Each transit router checks its output queues for deciding congestion levels as the local detection

✓ In global detection

- Identify an attack and its route



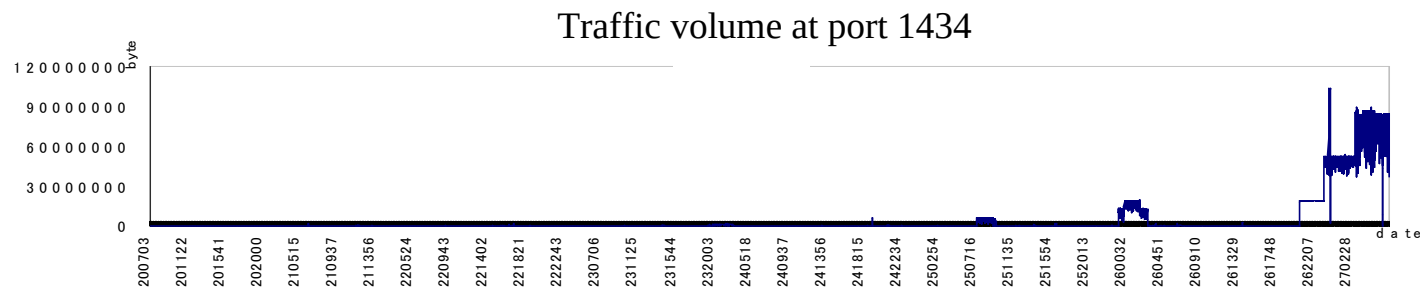
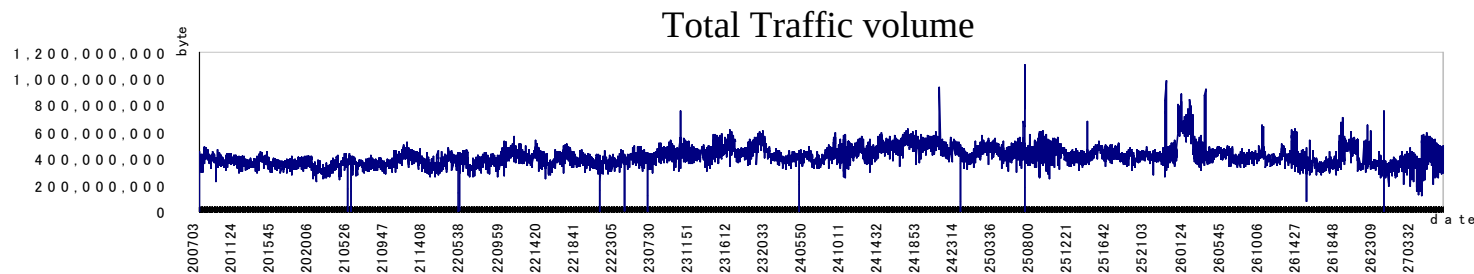
Intrusion Forecasting Method

(3) Time-Series Analysis

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■ Main Idea

- Detect intrusions early in broadband networks using the exponential smoothing method
- Extract traffic volume at a destination port to find anomalies earlier and more precisely

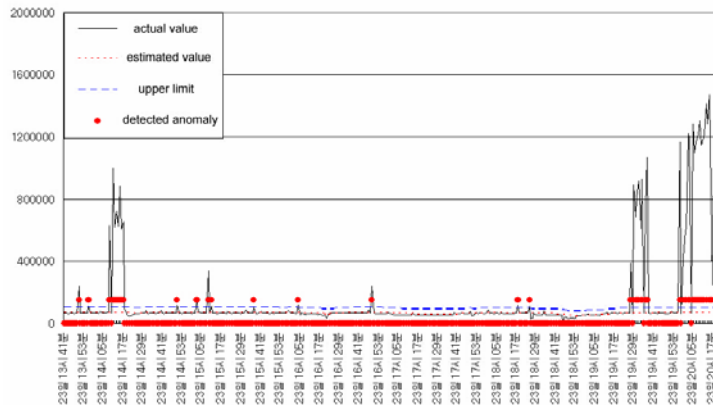


A CASE STUDY : Time-Series Analysis

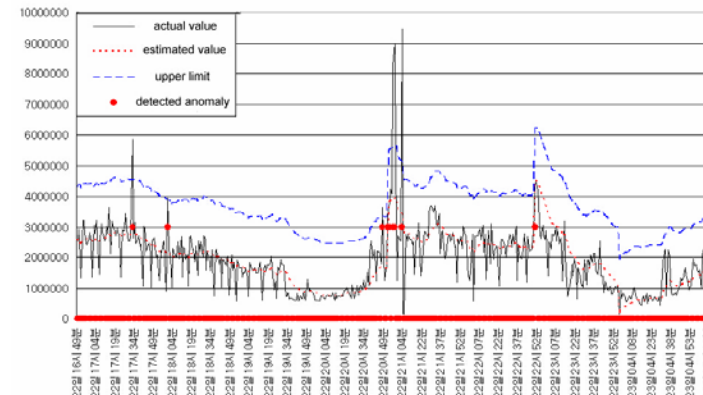
Fast detection scheme for broadband network using traffic analysis

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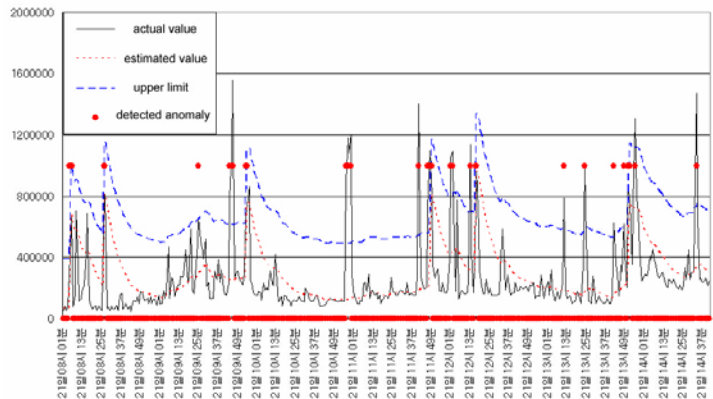
Experimental Results



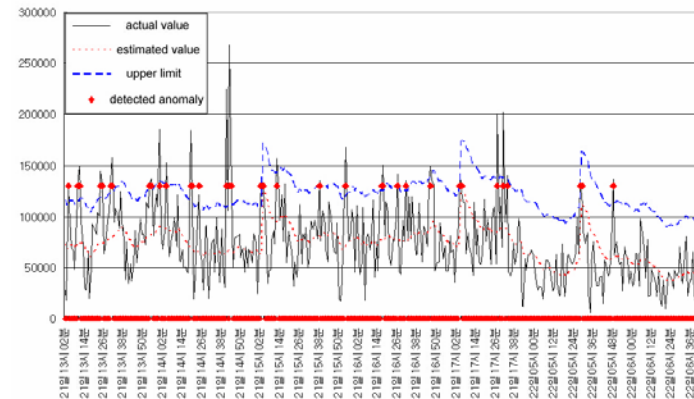
Detection of anomalies at port 1434



Detection of anomalies at port 80



Detection of anomalies at port 445



Detection of anomalies at port 137

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Conclusions

- **To defend networks against current cyber attacks, the importance of EWS is emphasized**
 - Many countries operate EWS through CERT
- **Intrusion forecast system**
 - Most significant part in EWS
 - Intrusion forecasting system architecture
 - ✓ Data Collection module
 - ✓ Data Analysis module
 - ✓ Reporting module
- **Intrusion forecasting techniques**
 - Forecasting in real-life
 - ✓ Weather, stock, power, etc.
 - Forecasting methods against cyber attacks
 - ✓ Data mining method
 - ✓ Probabilistic modeling
 - ✓ Time-series analysis

Thank You !